Compact Limits and a Conditional Maximal-Function Reduction for Kakeya-Type Carriers

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Abstract

This note records two elementary implications in an axiomatic Kakeya setting. First, a decreasing sequence of compact partial-direction carriers has a compact null intersection when the directions eventually exhaust the direction space and the carriers admit covers of vanishing Q-content. Second, an assumed L^p estimate for the associated Kakeya maximal operator gives a lower bound for the volume of neighbourhoods of any carrier and hence a lower Minkowski-dimension bound. An arbitrarily small power loss yields full lower Minkowski dimension. The maximal estimate is an external hypothesis; no such estimate is proved here. Consequently, the Euclidean statement below is a conditional reduction and does not constitute a proof of the Euclidean Kakeya conjecture.

1 Scope and notation

The purpose is limited to separating a compactness argument from the maximal-function input. Axiomatic treatments of Kakeya-type sets in metric spaces are developed in substantially greater depth by Venieri [1]; standard background on dimension and geometric measure theory may be found in Mattila [2]. The present note claims no new maximal estimate or incidence theorem.

Let (X, d) be a proper metric space and let μ be a locally finite Borel measure with $X = \operatorname{supp}\mu$. Assume that for some $Q > 0$, $r_0 > 0$, and constants $0 < c_{\mathrm{A}} \le C_{\mathrm{A}} < \infty$,

$$c_{\mathrm{A}} r^Q \le \mu(B(x,r)) \le C_{\mathrm{A}} r^Q \qquad (x \in X,\ 0 < r \le r_0). \tag{1.1}$$

Thus (X, d, μ) is Q-Ahlfors regular on the scales under consideration.

Let (Y, ρ, ν) be a compact metric probability space of directions, let A be a compact metric parameter space, and let

$$\Gamma : Y \times A \times [0,1] \longrightarrow X$$

be continuous. For $u \in Y$ and $a \in A$, define

$$F_u(a) := \{\Gamma(u,a,t) : 0 \le t \le 1\}, \qquad T_\delta(u,a) := \{x \in X : \operatorname{dist}(x, F_u(a)) \le \delta\}. \tag{1.2}$$

For $E \subset Y$, call $K \subset X$ an E-carrier if for every $u \in E$ there is $a \in A$ such that $F_u(a) \subset K$. A Y-carrier will simply be called a carrier.

Assume that $0 < \mu(T_\delta(u,a)) < \infty$ for sufficiently small δ, and that

$$\mathcal{M}_\delta f(u) := \sup_{a \in A} \frac{1}{\mu(T_\delta(u,a))} \int_{T_\delta(u,a)} |f| \, \mathrm{d}\mu \tag{1.3}$$

has a ν-measurable representative for the functions used below. For a nonempty bounded $K \subset X$, write

$$K_\delta := \{x \in X : \operatorname{dist}(x,K) \le \delta\}$$

and use the volume definition

$$\underline{\dim}_{\mathrm{M}}^{\mu} K := Q - \limsup_{\delta \downarrow 0} \frac{\log \mu(K_\delta)}{\log \delta}. \tag{1.4}$$

2 The two reductions

Proposition 2.1 (Compact null limit)**.** *Suppose that $Y_1 \subset Y_2 \subset \cdots \subset Y$ and $\bigcup_n Y_n = Y$. Let*

$$K_1 \supset K_2 \supset \cdots$$

be nonempty compact subsets of X, with K_n a Y_n-carrier. If each K_n admits a finite or countable cover

$$K_n \subset \bigcup_i B(x_{n,i}, r_{n,i}), \qquad 0 < r_{n,i} \le r_0, \qquad \sum_i r_{n,i}^Q \le \varepsilon_n, \quad \varepsilon_n \longrightarrow 0, \tag{2.1}$$

then

$$K := \bigcap_{n=1}^{\infty} K_n$$

is a nonempty compact carrier and $\mu(K) = 0$.

Proof. The nested nonempty compact sets K_n have the finite-intersection property, so K is nonempty and compact.

Fix $u \in Y$. Choose N with $u \in Y_N$. For every $n \ge N$, select $a_n \in A$ such that $F_u(a_n) \subset K_n$. Compactness of A gives a subsequence $a_{n_k} \to a \in A$. If $m \ge 1$ and $t \in [0,1]$, then for all sufficiently large k,

$$\Gamma(u, a_{n_k}, t) \in K_{n_k} \subset K_m.$$

Continuity of Γ and closedness of K_m imply $\Gamma(u, a, t) \in K_m$. Since m and t were arbitrary, $F_u(a) \subset K$. Thus K is a carrier.

Finally, equations (1.1) and (2.1) and countable subadditivity give

$$0 \le \mu(K) \le \mu(K_n) \le C_A \sum_i r_{n,i}^Q \le C_A \varepsilon_n.$$

Letting $n \to \infty$ proves $\mu(K) = 0$. □

Proposition 2.2 (Maximal estimate implies a dimension bound)**.** *Let $K \subset X$ be a bounded carrier. Suppose that for some $1 \le p < \infty$, $\beta \ge 0$, and $C < \infty$,*

$$\|\mathcal{M}_\delta f\|_{L^p(Y,\nu)} \le C\delta^{-\beta} \|f\|_{L^p(X,\mu)} \tag{2.2}$$

for all sufficiently small $\delta > 0$. Then

$$\mu(K_\delta) \ge C^{-p} \delta^{p\beta}, \qquad \underline{\dim}_M^\mu K \ge Q - p\beta. \tag{2.3}$$

Proof. For each $u \in Y$, choose $a(u) \in A$ with $F_u(a(u)) \subset K$. Then $T_\delta(u, a(u)) \subset K_\delta$. With $f = \mathbf{1}_{K_\delta}$, this tube has average one, while every tube average is at most one. Hence

$$\mathcal{M}_\delta \mathbf{1}_{K_\delta}(u) = 1 \qquad (u \in Y).$$

Since $\nu(Y) = 1$, equation (2.2) yields

$$1 \le C\delta^{-\beta} \mu(K_\delta)^{1/p},$$

which is the volume bound in equation (2.3). Taking logarithms, dividing by $\log \delta < 0$, and letting $\delta \downarrow 0$ gives

$$\limsup_{\delta \downarrow 0} \frac{\log \mu(K_\delta)}{\log \delta} \le p\beta.$$

Substitution into equation (1.4) proves the dimension estimate. □

Corollary 2.3 (Conditional full dimension)**.** *Assume the hypotheses of proposition 2.1. If for one $1 \le p < \infty$ and every $\eta > 0$ there is $C_\eta < \infty$ such that*

$$\|\mathcal{M}_\delta f\|_{L^p(Y,\nu)} \le C_\eta \delta^{-\eta} \|f\|_{L^p(X,\mu)} \tag{2.4}$$

for all sufficiently small δ, then $K = \bigcap_n K_n$ is a compact null carrier and

$$\underline{\dim}_M^\mu K = Q.$$

Proof. Proposition 2.1 gives the compact null carrier. Applying proposition 2.2 with $\beta = \eta$ gives $\underline{\dim}_M^\mu K \ge Q - p\eta$ for every $\eta > 0$. Letting $\eta \downarrow 0$ gives the lower bound Q; the ambient Ahlfors-regular dimension gives the reverse bound. □

3 Euclidean specialization

Let $X = \mathbb{R}^d$, let $\mu = \mathcal{L}^d$, let $Y = S^{d-1}$ carry normalized spherical measure, and put

$$L_u(a) := \left\{ a + \left(t - \frac{1}{2} \right) u : 0 \le t \le 1 \right\}.$$

Write $T^{\mathrm{E}}_\delta(u,a)$ for the closed δ-neighbourhood of $L_u(a)$ and $v_{d,\delta} := \mathcal{L}^d(T^{\mathrm{E}}_\delta(u,a))$, which is independent of (u,a). Define the global Euclidean operator

$$\mathcal{K}_\delta f(u) := \sup_{a \in \mathbb{R}^d} \frac{1}{v_{d,\delta}} \int_{T^{\mathrm{E}}_\delta(u,a)} |f(x)| \, \mathrm{d}x. \tag{3.1}$$

Corollary 3.1 (Euclidean conditional reduction)**.** *Let $Y_n \uparrow S^{d-1}$, and let $K_n \downarrow K$ be nonempty compact sets such that K_n contains a unit segment in every direction in Y_n. Suppose*

$$K_n \subset \bigcup_i B(x_{n,i}, r_{n,i}), \qquad \sum_i r^d_{n,i} \longrightarrow 0.$$

If for some $1 \le p < \infty$ and every $\eta > 0$,

$$\|\mathcal{K}_\delta f\|_{L^p(S^{d-1})} \le C_\eta \delta^{-\eta} \|f\|_{L^p(\mathbb{R}^d)} \tag{3.2}$$

for all sufficiently small δ, then K is compact, has Lebesgue measure zero, contains a unit segment in every direction, and

$$\underline{\dim}_{\mathrm{M}} K = d.$$

Proof. Take $A = K_1$ and $\Gamma(u,a,t) = a + (t - \frac{1}{2})u$. The midpoint of every segment contained in K_n lies in $K_n \subset K_1$, so the parameter space A contains all positions needed for the carrier condition. Lebesgue measure is d-Ahlfors regular, and the ball-cover hypothesis is precisely equation (2.1) with $Q = d$.

The maximal operator restricted to $a \in A$ is bounded pointwise by the global operator in equation (3.1). Therefore equation (3.2) supplies equation (2.4), and corollary 2.3 applies. □

Remark 3.2 (Interpretation)**.** The conclusion of corollary 3.1 is conditional on equation (3.2). The argument from that estimate to the dimension bound is the characteristic-function calculation in proposition 2.2. Establishing the estimate requires additional incidence or harmonic-analytic input beyond this note.

References

[1] L. Venieri, *Dimension estimates for Kakeya sets defined in an axiomatic setting*, arXiv:1703.03635, 2017.

[2] P. Mattila, *Fourier Analysis and Hausdorff Dimension*, Cambridge Studies in Advanced Mathematics 150, Cambridge University Press, 2015.